

Current Internet, Networking & Cloud IaaS

Lesson 62 — Modern Protocols and Cloud Responsibility

JKSSB Computer Awareness

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TCP: Ordered and Reliable Delivery

- ▶ **TCP** stands for **Transmission Control Protocol**.
- ▶ It is **connection-oriented**: the two ends set up a connection before any data moves.
- ▶ It guarantees **ordered** delivery: bytes arrive in the order they were sent.
- ▶ It guarantees **reliable** delivery: lost segments are detected and **retransmitted**.
- ▶ It provides **no timing guarantee**: delivery is correct and ordered, but not promised within a fixed time.

One line

TCP = ordered + reliable, but with no promise about **when**. (PYQ-18)

UDP: Fast but Unreliable

Point	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Ordering	Guaranteed order	No ordering guarantee
Reliability	Retransmits lost data	No retransmission
Speed	Slower (overhead)	Faster (light header)

UDP suits live voice, video, and games, where speed matters more than every byte arriving.

ICMP: Control Messages, Not Data Transfer

- ▶ **ICMP** stands for **Internet Control Message Protocol**.
- ▶ It carries **control and diagnostic** messages about the network itself.
- ▶ It reports **errors** (destination unreachable, time exceeded) and supports tools such as **ping** and **traceroute**.
- ▶ It has **no ports** and carries **no application data**: it does not move files, mail, or web pages.

Trap

ICMP never transfers application data. It only reports and diagnoses. (PYQ-18, PYQ-19, TRAP-19)

BGP: Routing Between Networks

- ▶ **BGP** stands for **Border Gateway Protocol**.
- ▶ It is the **inter-domain** routing protocol: it routes between **autonomous systems** (large independently managed networks).
- ▶ It decides the **paths between** ISPs and organisations on the Internet backbone.
- ▶ It is **not** a protocol for routing inside one small local network.

Trap

BGP = **between** networks, not within one LAN. Interior protocols such as OSPF and RIP work inside a domain. (PYQ-19, TRAP-20)

BGP vs Interior Routing

Point	BGP	OSPF / RIP
Scope	Inter-domain (between autonomous systems)	Intra-domain (inside one network)
Where used	Internet backbone, between ISPs	Inside one organisation or campus
What it answers	Which networks to cross	Which router to use next inside the network

Exam shortcut: the word “inter-domain” points to BGP.

IPv6: A Vast Address Space

- ▶ **IPv6** uses **128-bit** addresses (written as eight groups of hexadecimal digits).
- ▶ 128 bits give about 3.4×10^{38} addresses — vastly more than the 32-bit IPv4 space.
- ▶ The fixed IPv6 header is **40 bytes** long.
- ▶ This space is the reason IPv6 removes the need for address-sharing workarounds such as **NAT** (Network Address Translation).

Trap

IPv6 = **128-bit**, never 256-bit. (PYQ-18, TRAP-18)

IPv6 Has No Broadcast

- ▶ IPv4 supports three delivery styles: **unicast** (one receiver), **multicast** (a group), and **broadcast** (everyone on the network).
- ▶ IPv6 keeps **unicast**, **multicast**, and **anycast** (nearest of several receivers) — but it has **no broadcast**.
- ▶ The IPv6 replacement for a broadcast is a **multicast** to the “all-nodes” group.
- ▶ So “IPv6 broadcast address” is a **contradiction**: no such address exists.

Remember

IPv6 drops broadcast; use multicast instead. (PYQ-19)

HTTP/3 Runs on QUIC, Not TCP

- ▶ **HTTP** versions 1.1 and 2 run over **TCP**.
- ▶ **HTTP/3** is different: it uses **QUIC** (Quick UDP Internet Connections).
- ▶ **QUIC** runs over **UDP**, not over TCP, and builds reliability and security into itself.
- ▶ Result: faster connection setup and no head-of-line blocking across streams.

Trap

HTTP/3 uses **QUIC over UDP** — never TCP. (PYQ-19, TRAP-21)

DNS: Names to Addresses

- ▶ **DNS** stands for **Domain Name System**.
- ▶ It translates human-readable **domain names** (such as `example.org`) into numeric **IP addresses**.
- ▶ Resolution runs **hierarchically**: root servers, then top-level-domain servers, then the domain's own servers.
- ▶ DNS is a **directory lookup**, not a routing protocol and not a mail-transfer protocol.

One line

DNS answers “what number belongs to this name”.

Who Runs .org: PIR Under ICANN

- ▶ **.org** is a generic top-level domain, originally for non-profit organisations.
- ▶ The **.org** registry is operated by **PIR** (Public Interest Registry).
- ▶ **ICANN** (Internet Corporation for Assigned Names and Numbers) coordinates domain names and IP addresses at the global level.
- ▶ Keep the roles apart: **PIR** runs the .org registry; **ICANN** coordinates the naming system as a whole.

Remember

.org → **PIR**; global name/IP coordination → **ICANN**.

ARPANET: The First Packet Network

- ▶ **ARPANET** (Advanced Research Projects Agency Network) went live in **1969** and is the ancestor of the Internet.
- ▶ It proved that **packet switching** works across long distances.
- ▶ Its key ideas — decentralised routing and surviving the loss of individual links — became the design of the Internet.
- ▶ Later work on ARPANET produced the **TCP/IP** protocol family.

One line

ARPANET (1969) = the first working packet-switched network.

Packet Switching vs Circuit Switching

Point	Packet switching	Circuit switching
Data handling	Split into addressed packets	One dedicated circuit for the call
Path	Packets may take different routes	Fixed path held for the whole session
Efficiency	Shares links; robust to failure	Reserves capacity even when silent
Example	Internet, ARPANET	Traditional telephone call

The Internet uses packet switching; the old telephone network used circuit switching.

Router: Assertion–Reason Reading

- ▶ **Assertion (A):** A router connects **different networks** and forwards packets between them.
- ▶ **Reason (R):** A router reads the destination **IP address** and chooses the next hop from its **routing table**.
- ▶ Both A and R are **true**, and R is the **correct explanation** of A: reading the address is **how** the router forwards between networks.
- ▶ Do not confuse the router with the **switch** (forwards inside one network using MAC addresses) or the **hub** (repeats everything).

Worked reading

A office PC sends a packet to another branch: the **router** reads the destination IP and forwards it toward that network.

IaaS: Who Manages What

- ▶ **IaaS** stands for **Infrastructure as a Service**.
- ▶ The **provider** manages the physical infrastructure: **networking hardware**, servers, storage hardware, and virtualisation.
- ▶ The **customer** manages everything above that: the **operating system**, runtime, applications, and data.
- ▶ Exam anchor: in IaaS, the provider manages the **networking hardware**. (PYQ-20)

Trap

The IaaS provider does **not** manage your OS or your data. (TRAP-22)

IaaS vs PaaS vs SaaS

Model	Provider manages	Customer manages
IaaS	Hardware, networking, storage, virtualisation	OS, runtime, apps, data
PaaS	All of the above plus OS and runtime	Apps and data
SaaS	Everything up to the finished app	Only its own data and settings

Move down the list and the provider manages more.

Trap Alert: Near-Neighbour Distinctions

- ▶ TCP = **ordered + reliable**; UDP = **neither**. (PYQ-18)
- ▶ ICMP = **control messages only**, no data, no ports. (TRAP-19)
- ▶ BGP = **inter-domain** (between autonomous systems). (TRAP-20)
- ▶ IPv6 = **128-bit**, 40-byte header, **no broadcast**.
- ▶ HTTP/3 = **QUIC over UDP**, not TCP. (TRAP-21)
- ▶ .org → **PIR**; coordination → **ICANN**.
- ▶ IaaS provider = **networking hardware**; customer = **OS, apps, data**. (TRAP-22)

Lesson 62: Recall

- ▶ **TCP** guarantees order and reliability but no timing; **UDP** is connectionless and fast.
- ▶ **ICMP** carries control/diagnostic messages; it transfers no application data and uses no ports.
- ▶ **BGP** routes **between** autonomous systems; OSPF/RIP route inside one domain.
- ▶ **IPv6** = 128-bit addresses, 40-byte header, no broadcast (multicast replaces it); its space removes the need for NAT.
- ▶ **HTTP/3** uses **QUIC over UDP**, not TCP.
- ▶ **DNS** maps names to IP addresses; **PIR** runs .org under **ICANN** coordination.
- ▶ **ARPANET** (1969) proved **packet switching**, the basis of the Internet.
- ▶ A **router** joins networks by reading destination IPs; in **IaaS** the provider manages networking hardware while the customer manages OS, apps, and data.